CASE STUDY / PHARMACEUTICAL IT / NETWORK, COMPUTE & BACKUP

Connected operations. Protected business data.

Unique Pharmaceutical Laboratories

Division of J.B. Chemicals & Pharmaceuticals Ltd.
Kadaiya, Daman | Plant JB32

D-TECH infrastructure supply and installation across enterprise networking, Dell servers, desktop computing and external tape backup.

COMBINED ORDER VALUE: APPROX. USD 67,490

7 unique orders | 3 Dell servers | 22 desktop systems

Historical 2017-2019 engagement | Tax-inclusive value
Single FX basis: 05 November 2019; duplicates excluded.



Project scope | From workplace to application infrastructure

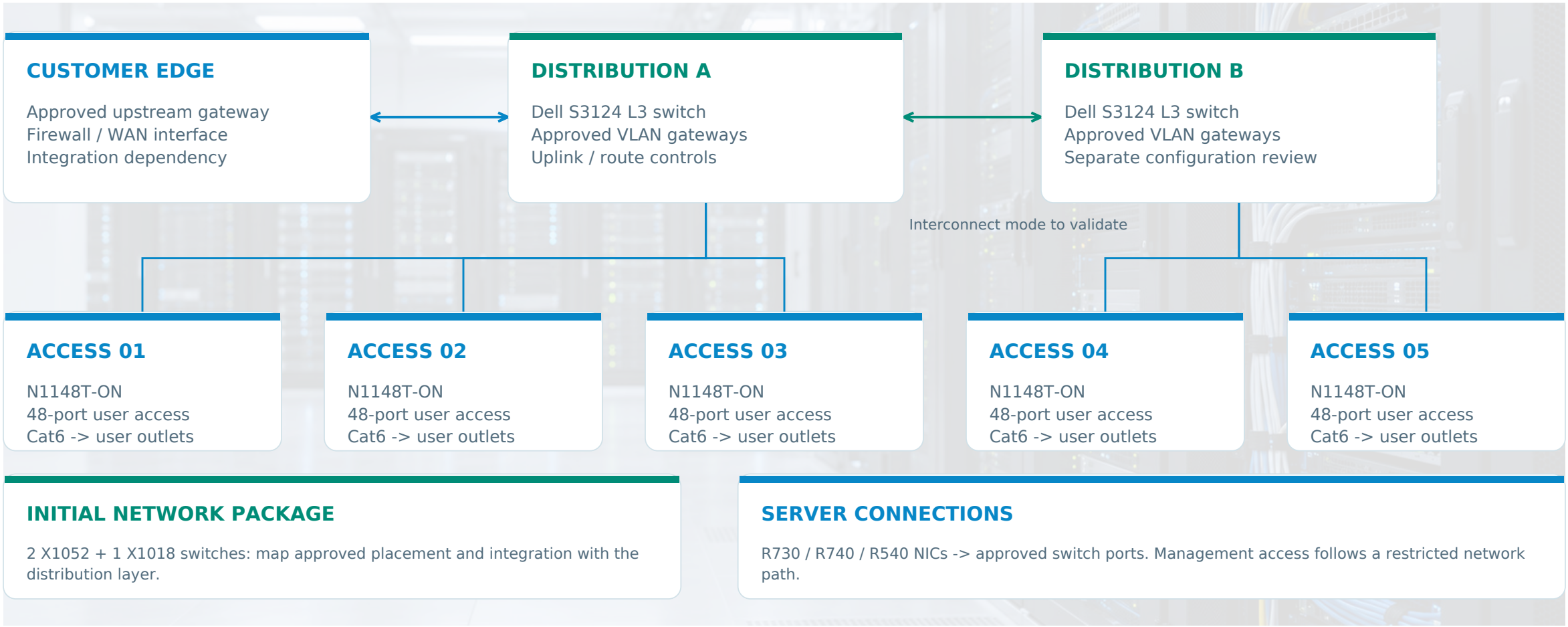
The pharmaceutical plant’s IT package spans workplace access, inter-area network connectivity, application-server capacity and recoverable backup. D-TECH brings these layers into one delivery approach: equipment selection and supply, structured installation, controlled configuration, functional testing and a usable handover.

Layer	Scope documented in the supplied orders	Purpose
Compute & users	PowerEdge R730, R740 and R540: 1 each; 22 OptiPlex 3050 desktops and 22 E1916H monitors. R740 PO specifically references PROPIX track & trace.	Application processing and user access.
Active network	2 S3124 L3 switches; 5 N1148T-ON L2 switches; 2 X1052 and 1 X1018 switches; optical modules.	Distribution, access and fibre-uplink interfaces.
Passive network	Cat6 outlets / panels / patching, single-mode fibre, LIUs and racks; a separate installation service order.	Traceable cable paths between users, racks and network equipment.
Storage & recovery	R540 internal SAS disk configuration; 1 external PowerVault LTO-6-200 tape drive in the initial package.	Local server storage and removable backup media workflow.

03 Network drawing | 04 Network implementation | 05 Server / disk architecture | 06 Tape backup & restore | 07 Delivery flow | 08 Customer care.

ILLUSTRATIVE LOGICAL DRAWING / DELL SWITCHING / FIBRE & COPPER LINKS

Network architecture | A clear path from users to servers



Network implementation | Micro-level connection and control



Work item	Implementation and verification
FIBRE & OPTICS	Verify connector, speed, wavelength, fibre type and optical budget; label cores / LIU ports, clean terminations and record link / fibre tests. Eight 10G single-mode optics are listed in the 2018 network expansion.
COPPER & PATCHING	Map switch port -> rack patch lead -> panel -> Cat6 permanent link -> jack -> user lead. Record outlet IDs, termination quality and cable certification against the approved standard.
LOGICAL CONTROLS	Define user, application and management segments; configure VLANs / trunks / gateways only on validated equipment. Confirm routing, access policy, loop protection and approved time / logging settings.
UPLINK CAPACITY	2018 scope includes five access switches but only eight 10G optics. A five-link optical design ordinarily needs ten endpoint optics; reconcile existing optics or alternate uplinks before assigning ports.

Two distribution switches do not establish stacking or automatic failover. Confirm hardware / software modes and test the approved design; no resilience measurements supplied.

Server storage | Inside the R540 data path

COMPUTE & MANAGEMENT

2 x Xeon Silver 4110
16 cores / 32 threads total
32 GB RAM | iDRAC9 Enterprise

APPLICATION / OS

Approved business workload
OS / platform licence to confirm
R540 order specifies no OS

PERC H730P

2 GB NV cache
11 ordered SAS disks
RAID supplied unconfigured

BOOT VOLUME | DESIGN OPTION

2 x 300 GB SAS -> RAID 1
Nominal usable: 300 GB
Mirror protects against one disk loss
Approve before initialisation

DATA VOLUME | DESIGN OPTION

8 x 600 GB SAS -> RAID 6
Nominal usable: 3.6 TB
Dual-parity design; performance
and workload suitability to review

SPARE & OPERATIONS

Remaining 1 x 600 GB -> hot spare
Record health, alerts and rebuilds
Capacity shown before formatting
RAID does not replace backup

Raw R540 capacity: 2 x 300 GB + 9 x 600 GB = 6.0 TB decimal. Layout is illustrative, not confirmed installed RAID. No shared SAN is listed.

R730 and R740 internal configurations are not supplied. Three server orders do not establish a hypervisor cluster, workload failover or shared storage.

Backup & recovery | Copy, protect and prove restoration

SOURCE DATA

Approved R730 / R740 / R540 applications and data
Inventory and ownership register

BACKUP HOST / SOFTWARE

Confirm licensed backup tooling
Application-consistent method
Compatible host adapter / cable

EXTERNAL TAPE DRIVE

PowerVault LTO-6-200
One external drive ordered
Validate interface and media

MEDIA CONTROL

Label cartridge / backup set
Secure custody and rotation
Retention / offsite handling according to customer policy

RESTORE VERIFICATION

Read selected backup to isolated test target; validate files or app recovery with the owner; record test results and elapsed time

RECOVERY READINESS

Maintain restore instructions
Approved RPO / RTO targets
Track failed jobs and repeat tests
No achieved RPO / RTO supplied

Data flow | direct host interface to tape, not an assumed Ethernet storage link

Drive supply does not prove installed backup software, completed jobs, purchased media or offsite retention. Confirm these in the operational handover.

Delivery discipline | Supply, install, test and hand over

01 | SURVEY & APPROVE

Map application users, switch / fibre routes, rack / power constraints and approved security controls.

02 | SUPPLY & INSPECT

Verify models, serials, disk / optic compatibility and quantities. Coordinate site delivery documentation.

03 | INSTALL & LABEL

Rack equipment, lay / terminate cables, splice fibre and label outlets, cores, panels and power feeds.

04 | CONFIGURE

Apply approved switch settings and server disk layout; prepare licensed OS and backup interfaces.

05 | TEST & ACCEPT

Test user-to-service paths, fibre links, server health, permissions and representative recovery checks.

06 | OPERATE & SUPPORT

Provide diagrams, configurations, asset / media records, runbooks and customer escalation contacts.

Coordinate production change windows and rollback with the plant owner. Infrastructure installation alone does not establish pharmaceutical system validation.

D-TECH value | One coordinated infrastructure partner

CUSTOMER & LOCATION

Unique Pharmaceutical Laboratories
Div. of J.B. Chemicals & Pharmaceuticals Ltd.
Survey No. 101/102, Airport Road, Kadaiya,
Nani Daman 396210 | Plant JB32.
Supplier named in historical POs: D Tech Computers.

WHY THIS ENGAGEMENT MATTERS

One coordinated scope connects end-user computing, structured cabling, switching, application servers and tape backup. D-TECH's delivery approach makes interfaces and service ownership understandable to the customer.

SEVEN UNIQUE ORDER REFERENCES | COUNTED ONCE EACH

4600060027 & 4700011332: 22 July 2017 | 4700011479: 11 September 2017 | 4700012082: 20 April 2018 | 4700012698: 17 November 2018 | 4700013590: 17 July 2019 | 4800162877: 05 November 2019.

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D-TECH Customer Care | support.dtechindia.com | +91 99989 03042

Value basis: seven unique tax-inclusive PO totals sum to INR 4,776,465.66. Both ZIPs contain identical copies of the same two orders; duplicates excluded.

FX: INR 70.7725 per USD, 05 November 2019 close, Exchangerates.org.uk. One presentation equivalent, rounded. Sources: supplied POs, logo and facility photo.